

DPEN021: Enabling Mathematics A

Class Questions Week 3

Week 3A

Key Idea - Solving Linear Equations

1. Solve the following equations:

(a) $11x + 17 = 39$	(b) $14 = -4 - 2t$	(c) $\frac{a}{-5} = -2$
(d) $\frac{q}{3} + 2 = 6$	(e) $7 = \frac{4-t}{3}$	(f) $5 - \frac{1-n}{2} = 4$
(g) $5 + 3x = 2x$	(h) $2 - 5t = 8 - 4t$	(i) $12y - 3y = 4y + 30$
(j) $4(x - 3) = 12$	(k) $4(p + 2) = -2p + 5$	(l) $7m - 11 = 6 - (15m - 9)$
(m) $8(3m + 2) = 4(5m + 10)$	(n) $-(y + 2) = 2(3y - 6)$	(o) $3(2b + 5) = 4 - (3b - 2)$
(p) $\frac{(d - 7)}{3} = 3d + 8$	(q) $9 - x = \frac{(4x - 17)}{-2}$	(r) $\frac{3(m - 2)}{4} + m = 10$

Key Idea - Solving Linear Equations with Fractions

1. Solve the following equations:

(a) $\frac{a}{3} = \frac{1}{2}$	(b) $\frac{2}{3} = \frac{b}{12}$	(c) $\frac{3x}{5} - \frac{x}{4} = 1$
(d) $\frac{x}{4} - 1 = \frac{x + 3}{2}$	(e) $\frac{5y}{4} - 3 = \frac{3y}{12}$	(f) $\frac{8p}{-7} = \frac{2p - 1}{5}$
(g) $\frac{x + 2}{2} = \frac{x - 3}{3}$	(h) $\frac{2x + 5}{3} = \frac{x + 2}{6}$	(i) $\frac{4 - 3m}{8} = \frac{2m + 1}{-4} + 2$
(j) $4 = \frac{10}{3k}$	(k) $\frac{5}{2b} - 2 = 1$	(l) $\frac{x}{x - 3} = 4$
(m) $\frac{3}{x - 1} = \frac{x}{x + 2} - 1$	(n) $\frac{3}{x + 1} = \frac{2}{x + 4}$	(o) $\frac{7}{2p} = \frac{3}{4p} - 4$
(p) $\frac{6}{x} + 2 = \frac{7}{2x} + 5$	(q) $\frac{1}{x} + \frac{3}{2x} - \frac{1}{3x} = 4$	(r) $\frac{3}{x - 1} + \frac{1}{x + 2} = \frac{1}{(x - 1)(x + 2)}$

Key Idea – Solving Quadratic Equations by Factorisation

1. Solve the following equations:

(a) $x^2 - 49 = 0$	(b) $5x^2 - 20 = 0$	(c) $4x^2 - 25 = 0$
(d) $x^2 - 16x = 0$	(e) $6x^2 + 9x = 0$	(f) $9x^2 + 36x = 0$
(g) $(x - 8)(x + 6) = 0$	(h) $3x(x + 4) = 0$	(i) $(2x - 1)(3x + 1) = 0$
(j) $x^2 - 4x - 21 = 0$	(k) $x^2 + 19x + 18 = 0$	(l) $x^2 - 11x + 24 = 0$
(m) $5x^2 + 6x - 8 = 0$	(n) $4x^2 + 12x + 9 = 0$	(o) $6x^2 + 11x - 35 = 0$
(p) $2x^2 = 13x - 20$	(q) $7x = 3(2 - x^2)$	(r) $-15 = x(8 + x)$

Week 3B

Key Idea – Quadratic Equations by Completing the Square

- Solve the following equations by completing the square, give your answer correct to 2 decimal places:
 - $x^2 + 6x = -4$
 - $x^2 - 4x = 3$
 - $x^2 - 10x = -6$
 - $x^2 + 8x - 7 = 0$
 - $x^2 - 7x + 4 = 0$
 - $x^2 + 5x - 8 = 0$
 - $4x^2 + 8x - 13 = 0$
 - $3x^2 - 6x + 2 = 0$
 - $2x^2 + 4x - 3 = 0$
- Solve the following equations by completing the square, give your answer in exact form:
 - $x^2 + 6x = 7$
 - $x^2 - 10x = -16$
 - $x^2 - 12x = -11$
 - $x^2 + 2x - 5 = 0$
 - $x^2 + 4x + 1 = 0$
 - $x^2 - 4x - 13 = 0$
 - $3x^2 - 5x + 2 = 0$
 - $2x^2 + 12x - 18 = 0$
 - $2x^2 + 3x - 5 = 0$

Key Idea - Solving Quadratic Equations using the Quadratic Formula

- Solve the following equations using the quadratic formula, give your answer correct to 2 decimal places:
 - $x^2 + 10x + 5 = 0$
 - $x^2 - 4x - 3 = 0$
 - $x^2 - 10x + 6 = 0$
 - $2x^2 + 3x - 6 = 0$
 - $x^2 + 6x - 10 = 0$
 - $4x^2 + 8x - 7 = 0$
 - $5x^2 = 4x + 3$
 - $7x - 9 + 6x^2 = 0$
 - $2 = 3x(2 - x)$
- Solve the following equations using the quadratic formula, give your answer in exact form:
 - $x^2 + 3x - 5 = 0$
 - $x^2 - 3x - 2 = 0$
 - $x^2 - x - 4 = 0$
 - $3x^2 + 7x + 1 = 0$
 - $3x - 9 + 2x^2 = 0$
 - $4x^2 - 9x + 4 = 0$
 - $x^2 = 2(3x + 1)$
 - $5x^2 + 3 = 10x$
 - $3 = 2x^2 + x$

Key Idea - Equations Reducible to Quadratic Form

- Solve the following equations by making suitable substitutions first:
 - $x^4 - 17x^2 + 16 = 0$
 - $x^4 - 10x^2 + 9 = 0$
 - $(x + 3)^2 + 4(x + 3) - 21 = 0$
 - $(3x - 4)^2 = 3x - 4$
 - $\left(x - \frac{5}{x}\right)^2 + 6\left(x - \frac{5}{x}\right) = 0$
 - $x^4 - 11x^2 + 18 = 0$
 - $(x^2 - 3x)^2 - 10(x^2 - 3x) + 21 = 0$
 - $(x - 1)^4 - 11(x - 1)^2 + 18 = 0$

Key Idea – Solving Equations with Powers

- Solve the following equations:
 - $6y^4 = 486$
 - $\frac{x^3}{3} = -9$
 - $\frac{2x^3 - 1}{5} = 3$
 - $y^{-5} = 32$
 - $2x^{-4} = 162$
 - $\frac{z^{-3} + 1}{2} = 14$
 - $2\sqrt{2 - 3x} = 8$
 - $\sqrt[3]{x - 2} = 5$
 - $\frac{\sqrt[4]{3x + 5}}{2} = 1$
 - $x^{\frac{2}{3}} = 16$
 - $z^{\frac{3}{2}} = 8$
 - $m^{\frac{4}{5}} = 81$
 - $2m^{\frac{3}{4}} = 250$
 - $x^{\frac{-2}{5}} = \frac{4}{9}$
 - $\frac{x^{\frac{5}{3}}}{4} = 8$
 - $\frac{a^{\frac{2}{3}}}{3} - 1 = 2$
 - $\frac{x^{\frac{4}{3}}}{4} + 2 = 6$
 - $\frac{2b^{\frac{5}{4}} - 1}{7} = 9$

Week 3C

Key Idea – Working with Formulas

1. $V = \pi r^2 h$ is the formula for the volume of a cylinder. Find h where $V = 180\pi$ and $r = 6$.
2. Given that the surface area of a cylinder is $A = 2\pi r h + 2\pi r^2$, where r is the radius and h is the height. Find h , given that $A = 20\text{cm}^2$ and $r = 1\text{ cm}$. Give your answer correct to one decimal place.
3. The velocity of a dropped object in m/s after falling a height h metres is given by the equation $v = \sqrt{2gh}$. Find the value of h when $v = 60\text{m/s}$, and $g = 9.8\text{m/s}^2$. Give your answer correct to two decimal places.
4. The law of a lifting jack is $E = \frac{C}{\eta} + D$. Given $E = 20$, $C = 5$ and $D = 2$, calculate η .
5. The electric current I flowing from a battery is given by $I = \frac{E}{R+r}$, where E is electromotive force, r is the internal resistance and R the external resistance. Make the internal resistance the subject.
6. Rearrange the formula of a circle $x^2 + y^2 = r^2$, to make y the subject.
7. Change the subject of each formula to the variable given:
 - (a) $y = mx + b$ [b]
 - (b) $A = \frac{1}{2}h(a+b)$ [a]
 - (c) $2y = \frac{x}{3} + 4$ [x]
 - (d) $s = ut + \frac{1}{2}at^2$ [a]
 - (e) $\frac{H}{h} = \frac{D}{d}$ [d]
 - (f) $A = 2ab^3 + 5$ [b]
 - (g) $2x = \sqrt{y} + 4$ [y]
 - (h) $H = \frac{2ab}{5} - 4c$ [a]
 - (i) $A = 180 - \frac{360}{n}$ [n]
 - (j) $V = \pi r^2 h$ [r]
 - (k) $x = \frac{1}{3}uv^2 + w$ [v]
 - (l) $A = 2xy^2 + 3x$ [y]
 - (m) $S = \frac{a}{1-r}$ [r]
 - (n) $P = \sqrt{\frac{2R-V}{5}}$ [V]
 - (o) $x = 2 - \sqrt{3y-1}$ [y]
 - (p) $\sqrt{\frac{1}{y} - \frac{1}{x}} = \frac{1}{4}$ [y]
 - (q) $m = \frac{2xy}{x+y}$ [y]
 - (r) $b = \frac{a+1}{a-1}$ [a]

Key Idea – Absolute Values

1. Evaluate the following:
 - (a) $|-7| + |4|$
 - (b) $|-3| \times |-5|$
 - (c) $|2-9| + |-8+3|$
 - (d) $|-2| \times |3|^2$
 - (e) $|6|^2 - 3|-5|$
 - (f) $|4-5|^2 \times |-2|$

Key Idea – Solving Absolute Value Equations

1. Solve the following equations, make sure you check your answers:
 - (a) $|x| = 4$
 - (b) $|2x| = 12$
 - (c) $|x-5| = 8$
 - (d) $|3x-2| = 10$
 - (e) $|x-4| + 3 = 8$
 - (f) $2|x-3| = 14$
 - (g) $|3x+6| = 2x+7$
 - (h) $|x-2| = 3x+4$
 - (i) $|2x+1| = x-5$
 - (j) $|2x-5| = |x+2|$
 - (k) $|3x-4| = |x+6|$
 - (l) $2|x+8| = 3|x+5|$

Week 3D

Key Idea – Solving Inequalities

1. Solve the following inequalities, write your answer in interval notation:

- | | | |
|--|--|---|
| (a) $-2 \leq x+1 \leq 0$ | (b) $x-5 \geq 2$ | (c) $-x > 4$ |
| (d) $6x > 9x+27$ | (e) $5 > -3x$ | (f) $3 \leq 3x+1 < 9$ |
| (g) $-1 < \frac{2x}{3} < 3$ | (h) $-21 < 3x \leq 6$ | (i) $-8 < 3x-2 < 4$ |
| (j) $5(x-3) \geq 2(x+7)$ | (k) $-1 < 2(x+4) \leq 0$ | (l) $-3(x-2) \leq 6(x+1)$ |
| (m) $\frac{1}{4} - \frac{x}{3} \leq 1$ | (n) $\frac{2}{3} \leq \frac{2x+3}{2} \leq 5$ | (o) $\frac{x}{4} \leq \frac{x}{6} + \frac{1}{12}$ |
| (p) $\frac{x+1}{2} - \frac{x-2}{3} \geq 7$ | (q) $\frac{x}{6} + \frac{2x+5}{3} < 4$ | (r) $\frac{x}{5} + \frac{3}{10} < \frac{x}{4}$ |

Key Idea – Solving Absolute Value Equations with Inequalities

1. Solve the following inequalities, write your answer in interval notation:

- | | | |
|--|--|--|
| (a) $ x > 5$ | (b) $ x \leq 4$ | (c) $ x \geq 6$ |
| (d) $ x+4 < 8$ | (e) $ x-7 \geq 2$ | (f) $ x+3 \leq 4$ |
| (g) $ 3x-2 \geq 10$ | (h) $ 2x+5 \leq 13$ | (i) $ 10x+6 \leq 12$ |
| (j) $\left \frac{x}{2}\right < 4$ | (k) $\left \frac{x}{2}-3\right > 5$ | (l) $\left \frac{2x}{5}\right \leq 3$ |
| (m) $\left \frac{3x-1}{2}\right \geq 1$ | (n) $\left \frac{2x-5}{3} + \frac{x}{6}\right \geq 4$ | (o) $\left \frac{x+3}{2} - \frac{x-1}{4}\right < 6$ |